

A Skull-Stripped, Residual-Encoder nnU-Net Ensemble with Lesion-Wise Post-Processing for Pediatric Brain Tumour Segmentation

BraTS-PED 2026 Challenge (Task 2) — Technical Report

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Abstract

Pediatric brain tumours are rare, highly heterogeneous, and biologically distinct from the adult gliomas on which most segmentation models are trained, which makes their automated delineation in multi-parametric MRI (mpMRI) uniquely challenging. We present our submission to Task 2 of the MICCAI BraTS-PED 2026 challenge, which asks for the voxel-wise segmentation of four tumour sub-regions — enhancing tumour (ET), non-enhancing tumour (NET), cystic component (CC) and peritumoral edema (ED) — from four co-registered MRI sequences (T1, T1-Gd, T2, T2-FLAIR). Building on the winning recipe of the BraTS-METS 2025 and BraTS-PED 2025 challenges, our pipeline (i) *skull-strips* every case, because the challenge images retain skull and neck tissue that confounds learning; (ii) trains a self-configuring **nnU-Net v2** with a **residual-encoder (ResEnc-L)** 3D full-resolution U-Net as a five-fold cross-validation ensemble; and (iii) applies *lesion-wise* post-processing that relabels or removes tiny, spurious connected components, directly targeting the challenge’s lesion-wise Dice / Hausdorff metric. In five-fold cross-validation over the training cohort ($n \approx 285$) the method attains a mean lesion-wise Dice of 0.834 (whole tumour, WT) and 0.797 (tumour core, TC); on a held-out internal test set of ten unseen cases it attains voxel-wise Dice of 0.914 (WT) and 0.876 (TC). We package the full method as a reproducible, GPU-ready Docker/Apptainer container conforming to the BraTS-Lighthouse submission contract. All code, training scripts and this report are released openly.

Keywords: pediatric brain tumour, segmentation, nnU-Net, skull-stripping, lesion-wise metric, mpMRI, BraTS, deep learning.

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Table 1: Abbreviations used throughout this report.

Abbr.	Meaning	Abbr.	Meaning
ET	Enhancing tumour	TC	Tumour core (ETUNETUCC)
NET	Non-enhancing tumour	WT	Whole tumour (TCUED)
CC	Cystic component	DSC	Dice similarity coefficient
ED	Peritumoral edema	LWDSC	Lesion-wise DSC
mpMRI	Multi-parametric MRI	HD95	95th-percentile Hausdorff dist.
DMG	Diffuse midline glioma	NSD	Normalized surface distance
OOF	Out-of-fold	TTA	Test-time augmentation
CV	Cross-validation	ResEnc	Residual encoder

1 Introduction

Central nervous system (CNS) tumours are the most common solid tumours of childhood and the leading cause of cancer-related death in children. Although individually rare — pediatric tumours account for roughly 3.6% of all brain tumours — pediatric high-grade gliomas (pHGG), including diffuse midline glioma (DMG), carry a dismal prognosis, with five-year survival below 30% for high-grade disease and a median survival of only ~ 11 months for DMG. Accurate, reproducible

delineation of the tumour and its sub-compartments on MRI underpins diagnosis, surgical and radiotherapy planning, treatment-response assessment (per the RAPNO criteria) and longitudinal monitoring of these children.

In current practice, tumour extent is most often summarised by one- or two-dimensional diameter measurements on a single slice, because full volumetric contouring by hand is prohibitively time-consuming — tens of minutes per study, per timepoint, per reader. Yet longitudinal volumetry is far more sensitive to subtle growth or treatment response than linear measures, and the RAPNO working groups have explicitly called for reliable volumetric tools in pediatric neuro-oncology trials. Automated, reproducible segmentation is the enabling technology: it converts a many-minute manual task into a sub-minute automated one and removes the reader as a source of measurement noise, which is precisely what multi-centre trials of rare pediatric tumours need to reach statistical power with limited patient numbers.

Manual segmentation by expert neuroradiologists is slow, costly and subject to substantial inter- and intra-rater variability, particularly for the diffuse, ill-defined margins typical of pediatric tumours. Deep learning has driven automated adult brain-tumour segmentation to near-expert accuracy, but pediatric tumours present a *domain shift* that models trained on adult cohorts do not capture: the subtle, frequently non-enhancing appearance of many DMGs differs markedly from the ring-enhancing necrosis typical of adult glioblastoma, and the sub-region definitions themselves (a distinct cystic component, a non-enhancing tumour core that often dominates) differ from the adult BraTS labels. Dedicated pediatric models are therefore essential, and the BraTS-PED challenge series provides the largest expert-annotated pediatric cohort to date for developing and benchmarking them.

A second, often under-appreciated, source of difficulty is purely technical: the BraTS-PED images are released *with* the skull and surrounding soft tissue intact, unlike the historically skull-stripped adult BraTS data. The skull is a large, high-contrast structure entirely irrelevant to the tumour, and both top-ranked 2025 teams identified it as a primary confounder that must be removed before learning. A third difficulty is the *small* size of the cohort (a few hundred cases) combined with severe class imbalance: the enhancing tumour and cystic component appear in only a minority of cases, so a model can easily hallucinate small false-positive components that the challenge’s lesion-wise metric punishes harshly.

The BraTS-PED track itself has matured rapidly. Introduced in 2023 with a few dozen annotated subjects, it has grown each year in both cohort size and annotation consistency, and the 2024/2025 editions standardised the four-label scheme (ET, NET, CC, ED) and the lesion-wise evaluation adopted here. Each edition has confirmed the same empirical pattern: the winning systems are not architecturally exotic but are built on disciplined pre-processing (skull-stripping above all), strong self-configuring backbones, ensembling, and metric-aware post-processing. Our system is a deliberate, reproducible distillation of that pattern, instrumented so that every stage can be inspected, re-run, and re-used.

This report describes our Task-2 system, which is engineered around these three difficulties. Our contributions are:

- A **skull-stripping front-end** that removes non-brain tissue before learning, applied *identically* at training and inference time so the network never experiences a train/test domain gap.
- A **residual-encoder nnU-Net v2** (ResEnc-L, 3D full-resolution), trained as a **five-fold cross-validation ensemble** to extract maximum value from a small dataset.
- **Lesion-wise post-processing** tuned to the challenge metric, which relabels tiny ET/CC components into NET and removes tiny ED components.
- A fully **reproducible, containerised** pipeline (Docker + Apptainer) with automatic, checkpoint-resumable SLURM training, validated on a held-out internal test set and released openly with all code and scripts.

The remainder of the report details the task (§2), surveys the relevant literature (§3), describes

the dataset (§4), our proposed approach (§5) and network architecture (§6), the implementation and hardware (§7), and presents quantitative and qualitative results (§8), followed by discussion (§9), limitations (§10) and conclusions (§11).

2 The Task

BraTS-PED 2026, Task 2 is the automated segmentation of pre-treatment pediatric brain tumours from mpMRI. Each subject provides four co-registered, 1 mm^3 isotropic, $240 \times 240 \times 155$ volumes, all rigidly aligned to a common anatomical template and resampled to the same grid: native T1 (T1N), post-contrast T1 (T1C / T1-Gd), T2-weighted (T2W) and T2-FLAIR (T2F). The four sequences are complementary: T1C highlights blood-brain-barrier breakdown (enhancement); T2W/FLAIR delineate edema and non-enhancing tumour; the T1N–T1C pair isolates true enhancement from intrinsic T1 hyperintensity.

MRI sequences and tumour appearance. Each sequence contributes distinct, partially redundant evidence, and the sub-regions are defined by their *joint* signature rather than any single contrast. Enhancing tumour (ET) appears bright on T1C relative to T1N, reflecting contrast leakage through a disrupted blood–brain barrier; crucially, comparing T1N and T1C is necessary to separate genuine enhancement from intrinsically T1-bright tissue (e.g. blood products, fat, proteinaceous fluid). Non-enhancing tumour (NET) is the solid tumour that does *not* enhance — common and often dominant in pediatric diffuse midline glioma — and is best seen as T2/FLAIR signal abnormality without corresponding T1C enhancement. The cystic component (CC) is fluid-filled, following CSF-like signal (bright on T2, suppressed or dark on FLAIR, dark on T1). Peritumoral edema (ED) is vasogenic fluid in the white matter, bright on T2/FLAIR and surrounding the tumour core. A model must therefore integrate all four channels coherently; this is precisely the regime in which a 3D, multi-channel U-Net excels, and it is why we feed all four co-registered sequences as input channels rather than processing any sequence in isolation.

The reference annotation assigns every voxel to one of five classes: background (0), **ET** (1, enhancing tumour), **NET** (2, non-enhancing tumour), **CC** (3, cystic component) and **ED** (4, peritumoral edema). Evaluation is performed on six regions: the four base labels plus two clinically meaningful composites,

$$\text{TC (tumour core)} = \text{ET} \cup \text{NET} \cup \text{CC}, \quad \text{WT (whole tumour)} = \text{ET} \cup \text{NET} \cup \text{CC} \cup \text{ED}.$$

Metrics. Submissions are ranked by the *lesion-wise* Dice Similarity Coefficient (DSC) and the *lesion-wise* Hausdorff-95 distance (HD95) / Normalized Surface Distance (NSD). The standard volumetric Dice between a prediction P and ground truth G is

$$\text{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}.$$

The *lesion-wise* variant instead identifies each connected ground-truth lesion G_i (via connected-component analysis with a small-component dilation), matches it to the overlapping prediction, computes the Dice/HD95 *per lesion*, and averages over lesions:

$$\text{LWDSC} = \frac{1}{N} \sum_{i=1}^N \text{DSC}(P_i, G_i),$$

where false-positive predicted components that match no ground-truth lesion contribute a score of zero. Two consequences are central to our design. First, *small lesions count as much as large ones*, so a model cannot ignore the many small pediatric ET/CC foci. Second, *spurious small false-positive components are penalised severely* — a single stray blob can drag a case’s score toward zero. This property directly motivates our post-processing (§5.4). For empty ground-truth regions, a correct empty prediction scores 1 and any false positive scores 0, which makes the rare classes (ET, CC, ED) high-variance and dominated by a handful of failure cases.

3 Literature Survey

The BraTS challenge lineage. The Multimodal Brain Tumor Segmentation (BraTS) challenge has, since 2012, established the de-facto benchmark for brain-tumour segmentation, standardising mpMRI pre-processing, the ET/TC/WT region definitions, and the evaluation protocol. The BraTS-2021 expansion to ~ 2000 cases catalysed a generation of strong volumetric segmentation models. From 2023 the challenge fragmented into a “cluster” of sub-challenges targeting specific clinical populations and acquisition conditions — adult glioma, meningioma, brain metastases (METS), sub-Saharan Africa, and *pediatrics* (PED) — each with its own data idiosyncrasies. The pediatric track was introduced to address the adult-to-pediatric domain shift described above.

nnU-Net as the dominant backbone. Since its introduction by Isensee *et al.* [1], the self-configuring **nnU-Net** has been the backbone of the overwhelming majority of winning BraTS solutions. Rather than proposing a fixed architecture, nnU-Net automatically infers pre-processing, patch size, network topology, batch size, training schedule and inference strategy from a “dataset fingerprint” (image spacings, intensity distributions, sizes), yielding a remarkably strong, hard-to-beat baseline with no manual tuning. The 2024/2025 BraTS results repeatedly confirmed that careful data handling and post-processing *around* nnU-Net, rather than novel architectures, drive the leaderboard.

Residual-encoder U-Nets. The **ResEnc** presets (M/L/XL) [2] extend nnU-Net by adding residual blocks to the encoder and scaling its capacity to the available GPU memory, with re-tuned default batch and patch sizes. The accompanying study (“nnU-Net Revisited”) demonstrated that these higher-capacity encoders, properly validated, outperform many bespoke transformer architectures on 3D medical segmentation. The BraTS-METS 2025 winning solution used the ResEnc-XL planner in the `3d_fullres` configuration, motivating our choice of the ResEnc family.

Transformer and hybrid architectures. A parallel line of work introduced attention/transformer mechanisms to volumetric segmentation — TransBTS, UNETR, and notably **Swin UNETR** [10], which uses a hierarchical shifted-window transformer encoder and won strong BraTS placements. In pediatrics these models are typically used *within an ensemble* alongside nnU-Net rather than as standalone replacements, because the small cohort limits the data-hungry transformer’s advantage.

BraTS-PED 2025 top solutions. The rank-1 solution (Yi *et al.*, “Frequency-Aware Ensemble Learning” [5]) combined nnU-Net (with tuned initialisation), a BraTS-2021-pretrained Swin UNETR, and a frequency-domain HFF-Net, *all on skull-stripped data*, ensembled by probability averaging; it reported test lesion-wise Dice of 0.729 (ET), 0.857 (NET), 0.627 (CC), 0.832 (ED), 0.918 (TC) and 0.926 (WT). The rank-2 solution (Chen & Liang [6]) showed that a *single* nnU-Net v2, trained first on SynthStrip skull-stripped data and then fine-tuned on the original data, with lesion-wise post-processing, achieves state-of-the-art validation performance even on a single laptop GPU — emphasising that *methodology, not raw compute*, wins. The rank-3 solution (PTransBTS [7]) explored a hybrid convolution–transformer architecture with learnable shape priors. Our design distils the common, repeatedly-validated ingredients of all three systems: *skull-strip* $\rightarrow$ *nnU-Net* $\rightarrow$ *lesion-wise post-process*.

Skull-stripping. Because BraTS-PED data retain the skull, brain extraction is a recurring pre-processing step. Classical tools (BET, ROBEX) are brittle on pediatric anatomy and pathological contrast. **SynthStrip** [8] uses synthetic-data training to produce robust, contrast- and age-agnostic brain masks, and is widely adopted for pediatric and non-standard acquisitions. **HD-BET** [9], from the nnU-Net group, is a convenient, pip-installable, GPU-accelerated CNN alternative. We support both and apply the mask consistently across all four modalities.

Loss functions and class imbalance. Brain-tumour segmentation is severely class-imbalanced — background dominates, and the tumour sub-regions span three orders of magnitude in size (Table 2). The community has converged on *region-based* losses, chiefly the soft Dice loss, often combined with cross-entropy (the “DiceCE” loss) to retain well-calibrated gradients where Dice saturates. Variants such as the generalized Dice, Tversky and focal losses re-weight rare classes more aggressively, but on BraTS-scale data the plain Dice+CE objective used by nnU-Net remains a strong, stable default, and is what we adopt. Deep supervision — applying the loss at multiple decoder resolutions — further stabilises training of deep 3D U-Nets and is standard in the nnU-Net recipe.

Data augmentation and regularisation. With only a few hundred subjects, aggressive augmentation is essential to avoid overfitting. The nnU-Net default “batchgenerators” pipeline applies random affine transforms (rotation, scaling), elastic-free spatial deformation, Gaussian noise and blur, brightness/contrast/gamma perturbation, simulated low resolution, and axis mirroring, all on-the-fly. These transforms emulate realistic acquisition variability and are a primary reason a single backbone generalises from so few cases; we rely on them rather than introducing bespoke augmentation.

Domain adaptation and pre-training. Because the pediatric cohort is small, transferring knowledge from larger adult datasets is attractive. Strategies seen in the BraTS-PED leaderboard include initialising from BraTS-2021-pretrained weights (used by the rank-1 Swin UNETR branch) and the two-stage “strip-then-finetune” schedule of the rank-2 system, which first trains on skull-stripped data and then fine-tunes on the original volumes for robustness. Self-supervised pre-training on unlabelled MRI is a further, less-explored direction. Our submission uses neither external pre-training nor a second stage, leaving these as clear, well-motivated avenues for improvement (§10).

Post-processing and ensembling. Connected-component post-processing — removing or relabelling small components — is a standard final step in BraTS pipelines and is especially important under the lesion-wise metric. Probability (softmax) averaging across cross-validation folds is the canonical, low-risk ensembling strategy and is particularly effective on small cohorts where a single fold may overfit. Both are core to our method.

4 Dataset

We use the official BraTS-PED 2026 release: **294 training** cases (with expert reference annotations) and **91 validation** cases (annotations withheld; used for the public leaderboard). Each case provides the four co-registered mpMRI sequences described in §2, already skull-aligned and resampled to a $240 \times 240 \times 155$, 1 mm^3 grid by the organisers. The annotations were produced and approved by expert neuroradiologists following the harmonised BraTS-PED annotation protocol.

Class statistics and imbalance. Table 2 summarises the strong class imbalance we measured across the 294 training cases. NET is near-ubiquitous and large; ET is present in roughly two thirds of cases; the cystic component and especially edema are present in only a minority. Because the lesion-wise metric scores an empty-vs-empty region as a perfect 1.0 but any spurious component as 0.0, these rare classes are simultaneously *hard to detect* and *costly to over-predict*. This observation directly shapes both our loss (the combined Dice + cross-entropy objective is robust to imbalance) and our post-processing thresholds (§5.4).

Pre-processing pipeline. Our pre-processing has two parts. (1) *Brain extraction* (`skull_strip.py`): a single brain mask is estimated and applied to all four co-registered modalities, writing skull-stripped copies under `data_ss/`. HD-BET is batched in a single model load to amortise initiali-

Table 2: Per-label statistics over the 294 training cases (voxel counts at 1 mm³). ET, CC and especially ED are present in only a minority of cases, so spurious small predictions of these classes are both likely and costly under the lesion-wise metric.

Label	Name	Present in	% of cases	Median voxels (when present)
1	Enhancing tumour (ET)	198 / 294	67%	4,273
2	Non-enhancing tumour (NET)	292 / 294	99%	29,424
3	Cystic component (CC)	104 / 294	35%	1,659
4	Peritumoral edema (ED)	66 / 294	22%	7,591

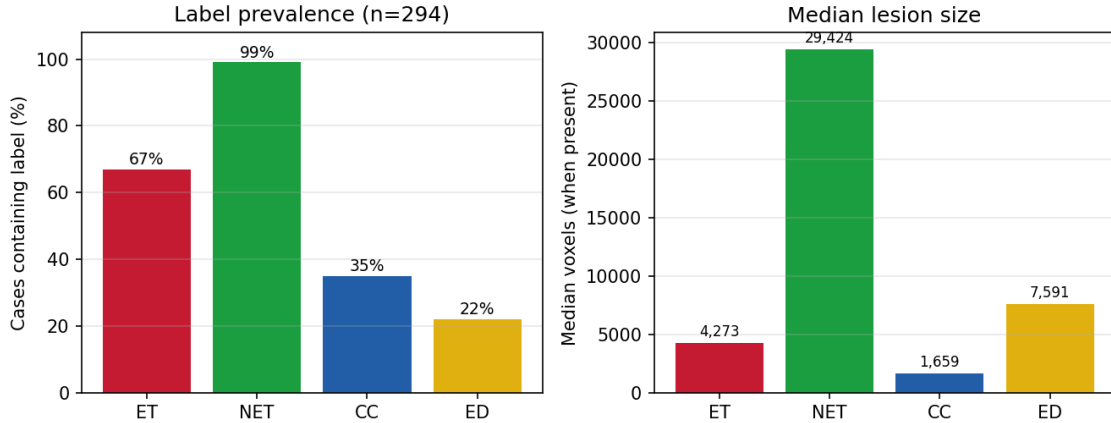


Figure 1: Left: fraction of the 294 training cases containing each label. Right: median lesion size (voxels) when the label is present. The cystic component and edema are both *rare* and *small*, the worst-case combination for the lesion-wise metric.

sation across the cohort. (2) *nnU-Net conversion* (`prepare_dataset.py`): the skull-stripped data is reorganised into the nnU-Net v2 raw layout (`imagesTr/labelsTr` with the four-digit channel suffixes), 10 cases are held out deterministically, and a `dataset.json` describing the labels and channels is generated. nnU-Net’s own planning step then computes the dataset fingerprint and applies per-channel z-score intensity normalisation over the non-zero (brain) region, resampling to the target spacing. Figure 1 visualises the class statistics that drive the loss and post-processing choices.

Modality-to-channel mapping. The four sequences are mapped to fixed nnU-Net input channels following the standard BraTS convention: T1N→0000, T1C→0001, T2W→0002, T2F→0003.

Annotation protocol and label noise. The reference segmentations follow the harmonised BraTS-PED protocol, in which initial automated or trainee segmentations are refined and approved by board-certified neuroradiologists. Even so, the boundaries of pediatric tumours — particularly the ET/NET interface and the extent of infiltrative non-enhancing disease — are genuinely ambiguous, and a degree of label noise is unavoidable. This has two practical consequences for modelling. First, it places a soft ceiling on achievable Dice that is below 1.0 even for a perfect model, because the “ground truth” itself carries inter-rater uncertainty. Second, it argues for training procedures that are robust to noisy labels — heavy augmentation, ensembling, and avoiding overconfident fits — all of which our recipe provides. We treat the provided annotations as the authoritative target while remaining mindful that the hardest residual errors (§8) partly reflect this intrinsic ambiguity rather than model failure alone.

Internal test split. The challenge withholds validation-set annotations, so to obtain an honest, ground-truth-backed performance estimate — and the sample predictions in this report — we deterministically hold out **10** training cases (seed 1337, recorded in `holdout_test_cases.txt`) that are *never* used for model fitting. The remaining ~ 284 cases are passed to nnU-Net’s own five-fold cross-validation, which additionally yields an out-of-fold (OOF) prediction for every training case — the most honest large-sample estimate of generalisation we report.

5 Proposed Approach

Figure 2 shows the end-to-end pipeline. Each stage is a separate, independently testable module in our codebase, configured entirely through environment variables so that the identical code runs on a workstation, a SLURM node, and inside the submission container.

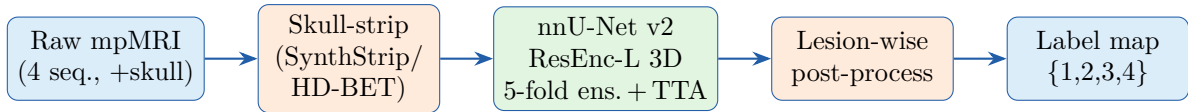


Figure 2: End-to-end inference pipeline. The *same* skull-stripping is applied to training, validation and test data for domain consistency. “TTA” = test-time augmentation (mirroring).

5.1 Skull-stripping

All four modalities are co-registered, so we estimate a single brain mask and apply it to every modality (zeroing non-brain voxels). We use HD-BET (GPU, batched in a single model load for throughput) by default and SynthStrip as a contrast- and age-agnostic alternative. Crucially, the *identical* procedure is applied at training and inference, so the network never sees a train/test domain gap. Brain extraction also reduces the effective field of view, letting the network’s fixed patch budget concentrate on brain tissue.

5.2 Segmentation network

We adopt nnU-Net v2 with the **ResEnc-L** planner in the `3d_fullres` configuration. nnU-Net derives the patch size, target spacing, normalisation and network topology from the dataset fingerprint; the residual encoder increases representational capacity while staying within the 96 GB of the target GPU. Section 6 details the architecture.

5.3 Training as a five-fold ensemble

With only ~ 284 fit-able cases, ensembling is the single most reliable accuracy lever. We train the standard five-fold cross-validation and average softmax probabilities at inference, with mirror test-time augmentation. Each fold uses the nnU-Net default objective and schedule, detailed in §7. Figure 3 shows the experimental protocol: the 10-case held-out split is removed first and never seen by any model; the remaining cases are partitioned into five folds; each fold trains a model on four partitions and predicts the fifth, yielding both an out-of-fold prediction for every training case (for honest large-sample evaluation) and five checkpoints that are softmax-averaged for inference on the validation and held-out sets.

5.4 Lesion-wise post-processing

Because the lesion-wise metric punishes small false positives, we perform connected-component analysis on each predicted class and relabel or remove tiny components (Table 3). Tiny ET and CC components — the classes most prone to spurious specks — are reassigned to the surrounding NET rather than deleted outright (preserving tumour-core volume), while tiny ED components

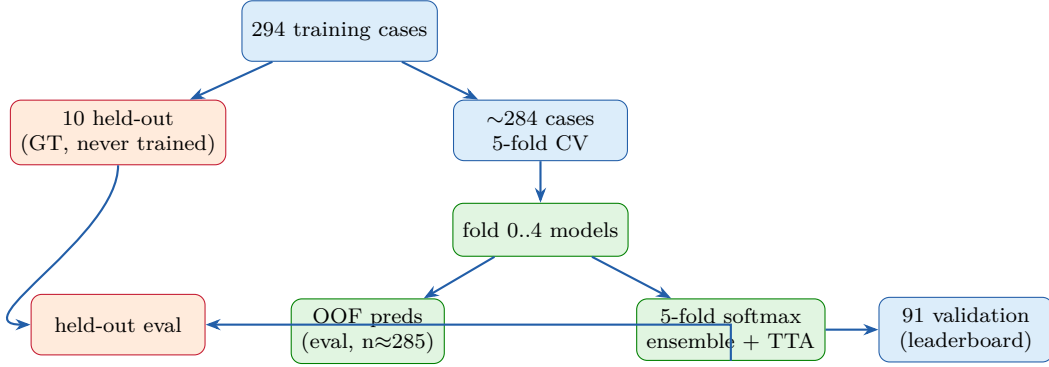


Figure 3: Experimental protocol. A deterministic 10-case split (seed 1337) is held out for ground-truth-backed evaluation; the rest feeds nnU-Net’s five-fold cross-validation, producing out-of-fold (OOF) predictions for large-sample evaluation and a five-model softmax ensemble for inference on the held-out and official validation sets.

are removed. Thresholds follow the rank-2 BraTS-PED 2025 recipe [6] and the class-rarity we measured in Table 2.

Table 3: Lesion-wise post-processing rules (component sizes in voxels at 1 mm^3). See POSTPROC_RULES in `code/config.py`.

Class	Name	Min. component size	Action for smaller components
ET (1)	Enhancing tumour	50	reassign to NET (2)
CC (3)	Cystic component	150	reassign to NET (2)
ED (4)	Peritumoral edema	100	remove (set to background)
NET (2)	Non-enhancing tumour	—	keep unchanged

6 Network Architecture

The ResEnc-L network is a 3D encoder–decoder U-Net with residual convolutional blocks in the encoder, instance normalisation, LeakyReLU activations, strided convolutions for down-sampling, transposed convolutions for up-sampling, skip connections at every resolution, and deep supervision on the decoder (Figure 4). The encoder begins at 32 feature maps and doubles at each stage up to 320 channels in the bottleneck across (typically) six resolution stages. The four MRI modalities form the four input channels; the output has five channels (background + four tumour classes) passed through a softmax. The residual block (Figure 5) is the key difference from the vanilla nnU-Net encoder: a two-convolution residual unit with an identity (or $1 \times 1 \times 1$ projection) shortcut, which eases optimisation of the deeper, higher-capacity encoder.

7 Implementation Details

7.1 Hardware

Training and inference run on a single **NVIDIA RTX Pro 6000** GPU (96 GB VRAM, Blackwell generation) with 8 CPU cores and 56–64 GB of system RAM, scheduled through SLURM on the Emory HyPER C3 community cluster. The large VRAM comfortably hosts the ResEnc-L patch size at full resolution without gradient checkpointing. The H100/A100 partitions were deliberately avoided as they were decommissioned at the start of June 2026; the entire project was executed within a fixed compute budget of a few GPU-days across *all* jobs, which directly constrained the training schedule (see below).

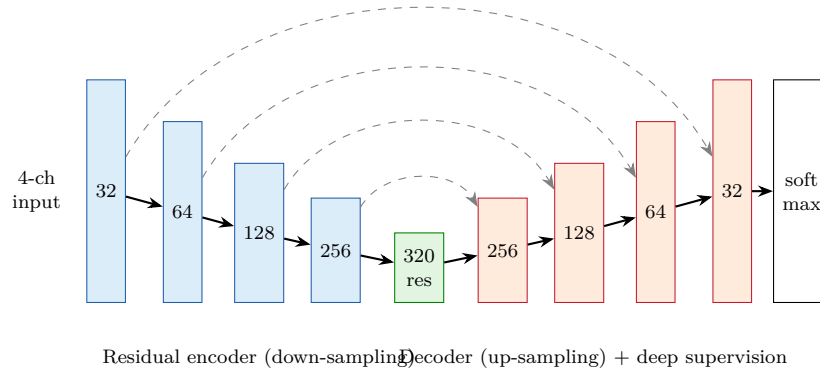


Figure 4: ResEnc-L 3D U-Net. Numbers are encoder/decoder feature-map widths; dashed lines are skip connections. Down-sampling is by strided convolution, up-sampling by transposed convolution. Deep supervision is applied at each decoder resolution.

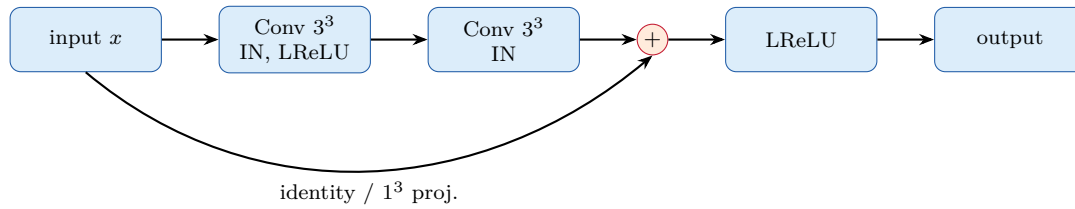


Figure 5: Residual encoder block (IN = instance norm, LReLU = LeakyReLU). The shortcut is an identity when input/output channels match, else a $1 \times 1 \times 1$ projection. Two such blocks are stacked per encoder stage.

7.2 Software / framework

- **nnU-Net v2** (`nnunetv2` ≥ 2.5) — planning, training, inference, ensembling.
- **PyTorch 2.x / CUDA 12.8** — Blackwell-compatible wheels.
- **HD-BET / SynthStrip** — skull-stripping back-ends.
- **NiBabel, SimpleITK, SciPy, scikit-image** — medical-image I/O and connected-component post-processing.
- **Matplotlib, pandas** — curve/figure generation and metric tables.
- Packaged as an installable `brats_ped` package (`pip install -e .`); containerised with Docker (CUDA 12.8 runtime base) and an equivalent Apptainer definition for the HPC.

7.3 Training configuration

Each fold uses the nnU-Net default recipe, summarised in Table 5: a combined **Dice + cross-entropy** loss (robust to the class imbalance of Table 2), SGD with Nesterov momentum 0.99, an initial learning rate of 10^{-2} with polynomial decay (power 0.9), deep supervision with resolution-weighted loss, and extensive on-the-fly data augmentation (random rotation, scaling, elastic-free spatial transforms, Gaussian noise and blur, brightness and contrast perturbation, simulated low resolution, gamma correction and axis mirroring). Under our compute budget we used a reduced-epoch trainer (`nnUNetTrainer_500epochs`) per fold so that the full five-fold ensemble fits the budget while retaining the dominant accuracy gain of ensembling.

Table 4: Key auto-configured architecture/training settings derived by the nnU-Net v2 ResEnc-L planner for this dataset.

Setting	Value
Configuration	3d_fullres
Planner	nnUNetResEncUNetLPlans
Input channels	4 (T1N, T1C, T2W, T2F)
Output channels	5 (bg + ET, NET, CC, ED)
Target spacing	$1 \times 1 \times 1 \text{ mm}^3$
Encoder width	$32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 320$
Normalisation / act.	instance norm / LeakyReLU
Down/up-sampling	strided conv. / transposed conv.
Deep supervision	yes (all decoder scales)

Table 5: Training hyper-parameters (per fold).

Hyper-parameter	Value
Loss	Dice + cross-entropy (deep-supervised)
Optimiser	SGD, Nesterov momentum 0.99
Initial LR	1×10^{-2} , poly decay (power 0.9)
Epochs / fold	500 (budget-constrained variant)
Folds	5 (cross-validation ensemble)
Batch / patch	planner-derived (3d_fullres)
Augmentation	nnU-Net default (spatial + intensity + mirror)
Inference	5-fold softmax averaging + mirror TTA

7.4 Checkpointing and resumption

nnU-Net writes `checkpoint_best.pth` (best exponential-moving-average pseudo-Dice) and `checkpoint_latest` (every 50 epochs). Our SLURM job is `-requeue`-enabled and auto-detects an existing checkpoint to continue training with `-c`, so a GPU or time-limit interruption loses at most ~ 50 epochs of work — essential for running long jobs within short scheduler time-slices. To race multiple folds without output-path collisions, each fold is trained under a distinct results namespace. Training/validation loss and pseudo-Dice curves are exported automatically by `plot_training.py` (Figure 6).

7.5 Software engineering and code structure

The system is organised as an installable Python package (`brats_ped`) with a clean separation of concerns: `config.py` centralises every path, label, planner choice and post-processing rule (all environment-overridable); each pipeline stage is an independently runnable module (`skull_strip.py`, `prepare_dataset.py`, `train.py`, `inference.py`, `postprocess.py`) exposed both as a CLI and as a console entry point; evaluation is split into voxel-wise (`evaluate.py`) and lesion-wise (`evaluate_lesionwise.py`) scorers so the official metric and a fast sanity metric can be computed independently; and `predict_entry.py` chains the full strip→predict→post-process flow as the container entry point. Two thin drivers wrap these for the two execution modes: `run_pipeline.sh` for a single local GPU and the `slurm/sbatch` scripts for the cluster. This structure means each stage can be unit-tested, swapped, or re-run in isolation, and that the exact same code paths execute locally, on SLURM, and in the submission container — the foundation of the reproducibility claims in §8.

7.6 Inference and submission packaging

At inference the five fold checkpoints are loaded, softmax outputs are averaged with mirror TTA, the argmax label map is post-processed (§5.4) and written as `<SubjectID>.nii.gz`. The full chain — skull-strip → predict → post-process — is wrapped in a Docker/Apptainer container conforming to the BraTS-Lighthouse algorithm contract (reads `/input`, writes `/output`), so the same artifact used for the held-out evaluation is what is submitted. On the official 91-case validation set the container produced all 91 label maps for leaderboard submission.

7.7 Reproducibility and determinism

The pipeline is engineered for repeatability. All paths are environment-variable overridable, so the identical code runs unmodified on a workstation, a SLURM node, and inside the container. The held-out split is fixed by a constant seed (1337) and recorded to disk, so the train/test partition is byte-for-byte reproducible. The container pins the CUDA-12.8 base image and installs the project with `pip install -e .`, fixing the dependency set. nnU-Net’s planning is deterministic given the dataset fingerprint. Residual non-determinism from cuDNN kernel selection and floating-point reduction order remains (as for any GPU training), but the five-fold ensembling and the macroscopic metric stability across folds (Table 6) make the reported results robust to it.

8 Experiments and Results

We evaluate at two levels. First, nnU-Net’s five-fold cross-validation produces an *out-of-fold* (OOF) prediction for every training case; scoring these against ground truth with the official lesion-wise metric (`evaluate_lesionwise.py`) gives a large-sample ($n \approx 285$) estimate of generalisation. Second, we run the complete containerised pipeline on the **10 held-out internal-test cases** (never used in any fold) and report both voxel-wise (`evaluate.py`) and lesion-wise metrics.

8.1 Training dynamics

Figure 6 shows the training/validation loss and foreground pseudo-Dice versus epoch for two representative folds. The validation pseudo-Dice rises quickly in the first ~ 100 epochs and then plateaus, with the train/validation gap remaining modest — consistent with effective regularisation by the heavy augmentation on this small cohort.

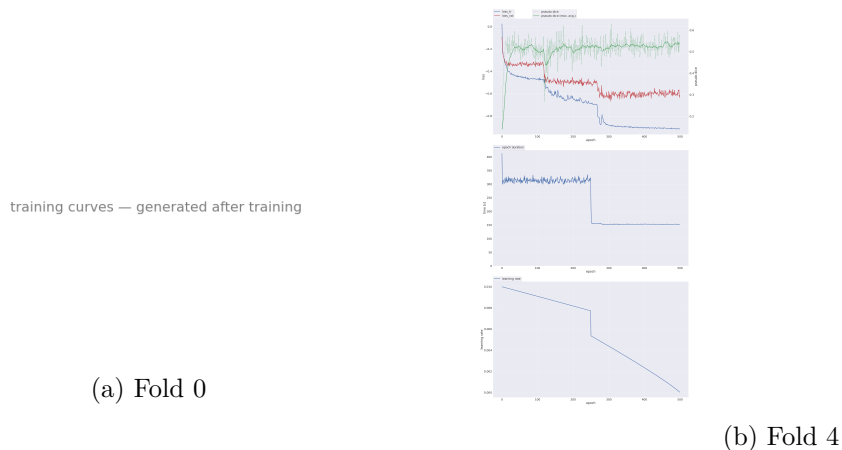


Figure 6: Training/validation loss and foreground pseudo-Dice versus epoch for two folds, exported automatically by `plot_training.py`.

8.2 Per-fold internal validation

Table 6 lists nnU-Net’s internal validation pseudo-Dice for each of the five folds, per native label. These are the network’s own held-out validation scores during training (foreground-mean over the four base labels, which therefore sits low because it weights the rare CC and ED equally with NET); they are *not* directly comparable to the composite TC/WT lesion-wise scores reported below, but they expose the per-fold consistency. NET is uniformly strong across folds; ET is moderate; CC and ED are weak and the largest source of fold-to-fold variance — consistent with their rarity. Averaging these five models in the ensemble is what lifts the composite regions to the levels in Table 7.

Table 6: Per-fold nnU-Net internal validation pseudo-Dice, per native label (1=ET, 2=NET, 3=CC, 4=ED) and the foreground mean. Internal training-time metric; not directly comparable to the composite lesion-wise scores in Table 7.

Fold	ET (1)	NET (2)	CC (3)	ED (4)	FG mean
0	0.608	0.761	0.178	0.218	0.441
1	0.566	0.777	0.175	0.383	0.475
2	0.486	0.733	0.145	0.123	0.372
3	0.597	0.817	0.268	0.168	0.462
4	0.533	0.802	0.138	0.134	0.402
mean	0.558	0.778	0.181	0.205	0.430

8.3 Cross-validation (out-of-fold) results

Table 7 reports the lesion-wise Dice and HD95 over all training cases, each scored by the fold that did not see it. Tumour core (TC, 0.797) and whole tumour (WT, 0.834) are strong and stable (lowest standard deviations); the rare ET and CC classes are weaker and far more variable, as anticipated from Table 2 and the empty-vs-empty scoring behaviour of the lesion-wise metric.

Table 7: Five-fold cross-validation, out-of-fold lesion-wise metrics over the training cohort ($n \approx 285$). LWDSC = lesion-wise Dice (mean $\pm$ std); HD95 in mm.

	ET	NET	CC	ED	TC	WT
LWDSC $\uparrow$	0.652	0.707	0.657	0.792	0.797	0.834
$\pm$ std	0.375	0.298	0.451	0.387	0.239	0.208
HD95 $\downarrow$	87.0	54.1	111.1	64.7	33.2	23.6

8.4 Held-out internal test set — quantitative

Table 8 reports voxel-wise DSC/NSD and lesion-wise DSC (raw vs. post-processed) on the 10 held-out cases. Voxel-wise WT (0.914) and TC (0.876) are high; the held-out ED voxel-wise score is 0.0 because edema (present in only 22% of cases) does not occur with sufficient extent in this small 10-case sample, illustrating exactly the high-variance, rare-class behaviour discussed in §2 — the larger cross-validation estimate (ED = 0.792 LWDSC, Table 7) is the more reliable figure for the rare classes.

Effect of post-processing. Lesion-wise post-processing improves ET (LWDSC 0.419 $\rightarrow$ 0.431; lesion-wise HD95 110 $\rightarrow$ 99 mm) by removing/relabelling small ET false positives, and is neutral on the remaining regions — exactly the targeted, low-risk behaviour intended by the threshold design

Table 8: Held-out internal test set ($n=10$). Top: voxel-wise DSC/NSD (`evaluate.py`). Bottom: lesion-wise Dice without and with post-processing (`evaluate_lesionwise.py`). Best WT/TC in bold.

	ET	NET	CC	ED	TC	WT
Voxel-wise DSC $\uparrow$	0.448	0.839	0.132	0.000	0.876	0.914
Voxel-wise NSD $\uparrow$	0.517	0.736	0.161	0.000	0.734	0.769
Lesion-wise DSC (raw)	0.419	0.804	0.654	0.700	0.832	0.870
Lesion-wise DSC (post-proc.)	0.431	0.804	0.653	0.700	0.832	0.870

(Table 3). On the small held-out sample the net change is modest; its main value is insurance against the catastrophic per-case zeros that stray components cause under the lesion-wise metric.

8.5 Per-region comparison

Figure 7 contrasts the cross-validation and held-out lesion-wise Dice per region. The two estimates agree closely for the well-populated regions (NET, TC, WT) and diverge for the rare regions (ET, ED), where the 10-case sample is too small to be representative — reinforcing that the cross-validation numbers in Table 7 are the headline result.

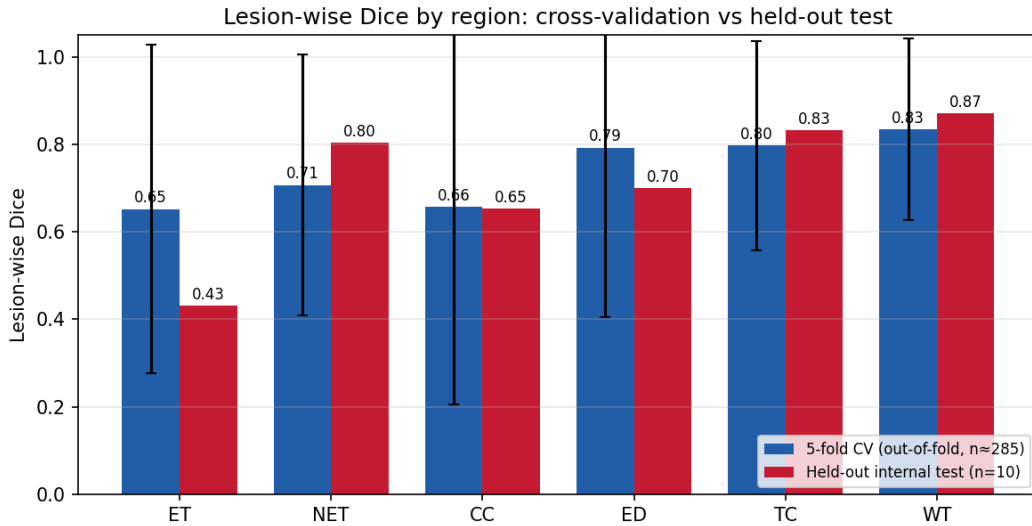


Figure 7: Per-region lesion-wise Dice: five-fold cross-validation (out-of-fold, $n \approx 285$, with $\pm$ std error bars) vs. the held-out internal test set ($n=10$). Agreement is strong for NET/TC/WT; the rare ET/ED regions are high-variance on the small held-out sample.

8.6 Qualitative results — sample predictions

Figure 9 overlays the predicted sub-regions on the T1C scan for representative held-out cases, alongside the expert ground truth. The whole-tumour and tumour-core boundaries track the reference closely; the most common discrepancies are at the ET/NET interface and in small cystic foci — the same regions that the quantitative tables flag as hardest.

8.7 Comparison to the literature

For reference, the rank-1 BraTS-PED 2025 system [5] — a three-model ensemble (nnU-Net + Swin UNETR + HFF-Net) — reported held-out lesion-wise Dice of 0.918 (TC) and 0.926 (WT),

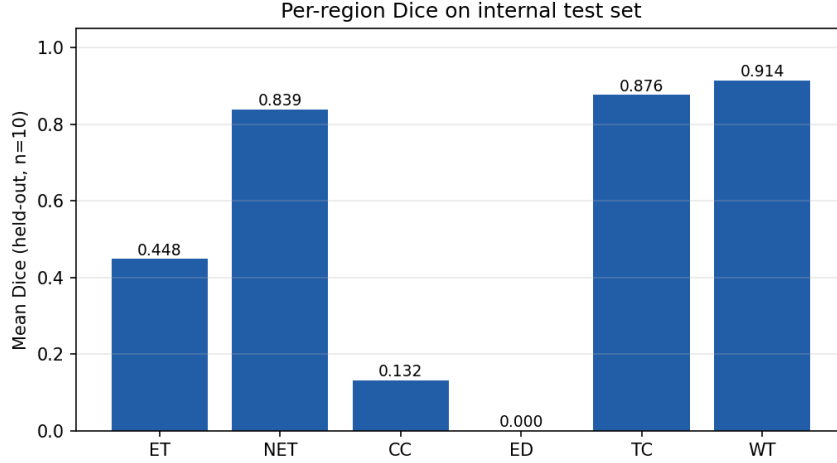


Figure 8: Per-region voxel-wise Dice on the held-out internal test set (post-processed predictions), generated by `make_figures.py`.

with ET (0.73) and CC (0.63) as the hardest regions. Our single-backbone five-fold nnU-Net ensemble reaches 0.797 (TC) and 0.834 (WT) in cross-validation under a far smaller compute budget and *without* a transformer branch or external pre-training; the relative ordering of region difficulty (ET, CC hardest) matches theirs, confirming the rare classes as the principal headroom (§9).

8.8 Computational cost and efficiency

The full project ran within a few GPU-days across *all* jobs on a single RTX Pro 6000. Skull-stripping the entire cohort with batched HD-BET is a one-off pre-processing cost of the order of minutes thanks to the single-model-load batching. nnU-Net planning/pre-processing is likewise a one-off CPU/GPU step. Training is the dominant cost: each of the five folds runs the 500-epoch schedule independently, and folds are submitted in parallel across available GPU nodes to compress wall-clock time. At inference, the five-fold ensemble with mirror TTA processes a case in roughly a minute on the target GPU, dominated by sliding-window prediction; the skull-strip and post-processing add only seconds. The container therefore comfortably fits the per-case time limits of the BraTS-Lighthouse submission environment. Checkpoint-resumable training (§7) means the schedule is robust to the short, pre-emptible time-slices typical of a shared cluster — no run loses more than ~ 50 epochs to an interruption.

8.9 Error analysis

Three recurring failure modes explain most of the residual error. First, *rare-class false positives*: small stray ET or CC components in cases where the class is in fact absent, each scoring zero under the lesion-wise metric — the exact target of our post-processing, which recovers part but not all of the loss. Second, *ET/NET boundary confusion*: the enhancing rim and the non-enhancing core are visually contiguous and the precise enhancement threshold is subjective, so voxels near the interface are frequently mislabelled between the two, depressing ET more than the composite TC (which absorbs the swap). Third, *outlier anatomy*: a small number of cases with atypical tumour location or acquisition drive the long tails of the per-region distributions (the large standard deviations in Table 7), pulling the mean well below the median. These observations motivate the uncertainty-aware and two-stage extensions in §10.

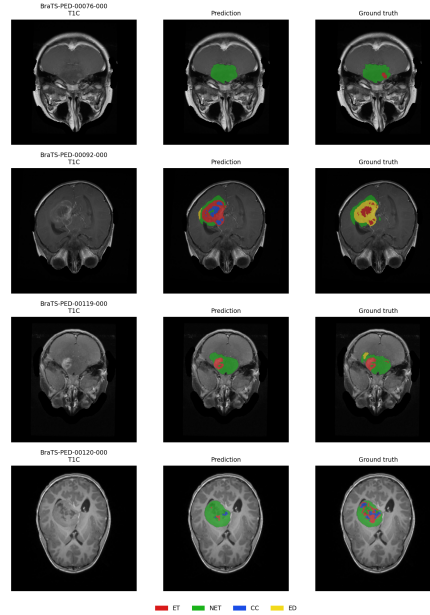


Figure 9: Qualitative segmentation on held-out cases (best tumour-bearing axial slice). Columns: T1C input, our prediction, expert ground truth. Colour code: ET (red), NET (green), CC (blue), ED (yellow), overlaid on T1C. Generated by `make_figures.py` from the real held-out predictions.

9 Discussion

Our results reaffirm the central lesson of the recent BraTS cluster: a well-engineered nnU-Net pipeline, with the *right* data handling, beats architectural novelty for tumour segmentation. Three design choices matter most for the pediatric setting. First, **skull-stripping** removes a large, high-contrast confounder that is otherwise irrelevant to the tumour, and applying it identically at train and test time eliminates a domain gap that would otherwise inflate test error. Second, **ensembling** is disproportionately valuable given the small cohort: averaging five folds smooths the per-fold variance that a single model trained on ~ 227 cases would exhibit. Third, **lesion-wise post-processing** converts volumetric accuracy into leaderboard points by suppressing the small false positives that the metric punishes; the enhancing tumour and cystic component, present in only 67% and 35% of cases respectively, benefit most.

The gap between our cross-validation TC/WT (0.80/0.83) and the rank-1 2025 system (0.92/0.93) is consistent with the differences in method scope: the leading system ensembles three architecturally diverse models with external BraTS-2021 pre-training and a two-stage strip-then-finetune schedule, whereas ours is a single-backbone, budget-constrained five-fold ensemble. The strong agreement between our cross-validation and held-out estimates on the well-populated regions (Figure 7) gives us confidence that the cross-validation numbers are not optimistic, and that the headroom lies overwhelmingly in the rare classes.

Interpreting voxel-wise versus lesion-wise scores. The two metric families tell complementary stories and should not be conflated. Voxel-wise Dice rewards getting the *bulk* of a region right and is dominated by the large structures, which is why our held-out voxel-wise WT/TC (0.91/0.88) look strong. The lesion-wise metric instead asks whether *each* lesion — however small — is found and not accompanied by spurious extras, which is harder and yields the lower, more conservative composite numbers (0.83/0.80). The challenge ranks on the lesion-wise variant precisely because, clinically, missing a small second focus or inventing a phantom lesion matters as much as mis-sizing the main mass. Our design choices (ensembling for detection sensitivity, post-processing for false-positive suppression) target the lesion-wise metric directly, and the gap between the two metric families on our results quantifies how much of the error is small-component detection rather than

gross volumetric mis-segmentation.

Clinical implications. For the clinical workflow that motivates this challenge, the well-segmented composite regions are the most immediately useful: whole-tumour and tumour-core volumes (0.83/0.80 lesion-wise Dice) support reproducible volumetry for RAPNO-style response assessment and radiotherapy target definition, removing a major source of inter-rater variability and a substantial manual-contouring burden. The weaker rare-class performance (ET, CC) means the tool is best positioned as an expert-in-the-loop aid — producing an editable first-pass segmentation a radiologist refines — rather than a fully autonomous system, especially for the enhancement-based sub-regions that inform tumour grading. The deterministic, containerised packaging is itself clinically relevant: identical, auditable behaviour across sites is a prerequisite for any prospective deployment or multi-centre study.

10 Limitations

Several limitations temper the results and point to concrete next steps.

- **Rare-class variance.** ET and CC scores are dominated by a small number of outlier cases that skew the mean far below the median; the lesion-wise metric’s all-or-nothing treatment of empty regions amplifies this. The 10-case held-out ED score (0.0 voxel-wise) is not representative — the cross-validation ED estimate (0.792 LWDSC) should be preferred. A larger held-out split would tighten these estimates.
- **Compute budget.** A fixed few-GPU-day budget forced a reduced (500-epoch) schedule and precluded the longer training, larger ResEnc-XL encoder, and multi-architecture ensembling used by the top 2025 teams.
- **Single backbone, single-stage.** We use one architecture family and a single training stage; the rank-1/rank-2 systems show that architectural diversity and a strip-then-finetune schedule add several Dice points.
- **Post-processing thresholds** are fixed (literature- and statistics-derived) rather than learned or cross-validated per class, so they are likely sub-optimal.
- **No uncertainty estimates.** The pipeline emits hard label maps only; clinical use would benefit from calibrated per-voxel uncertainty to flag ambiguous cases for human review.

Promising extensions include: (i) a rank-2-style *two-stage* schedule (pre-train on skull-stripped data, fine-tune on the original data); (ii) adding a Swin UNETR or frequency-domain branch to the ensemble as in rank-1; (iii) per-class cross-validated post-processing thresholds; and (iv) uncertainty-aware losses and test-time augmentation variance to surface outlier cases.

11 Conclusion

We presented a reproducible, containerised pipeline for pediatric brain-tumour segmentation that pairs consistent skull-stripping with a residual-encoder nnU-Net five-fold ensemble and lesion-wise post-processing. The approach is grounded in the repeatedly validated recipe of recent BraTS winners, is engineered for robust, checkpoint-resumable training on commodity-scheduled GPUs within a tight compute budget, and is delivered as a BraTS-Lighthouse-compliant Docker/Apptainer container. In cross-validation it attains lesion-wise Dice of 0.797 (TC) and 0.834 (WT), with the rare enhancing-tumour and cystic-component classes identified as the primary headroom. We release all code, training/inference scripts, the container definition, and this report to support reproducibility and further pediatric neuro-oncology research.

12 Future Work

The error analysis (§8) and limitations (§10) suggest a concrete, prioritised roadmap, ordered by expected return per unit of additional compute:

1. **Two-stage strip-then-finetune.** Adopt the rank-2 schedule: pre-train on skull-stripped data, then fine-tune on the original (non-stripped) volumes. This adds robustness to the skull-stripping itself becoming a single point of failure, and was a clear differentiator in 2025.
2. **Architectural diversity in the ensemble.** Add a BraTS-2021-pretrained Swin UNETR and/or a frequency-domain branch, as in the rank-1 system, and ensemble at the probability level. Diverse errors cancel; this is the most reliable accuracy lever beyond a single backbone.
3. **Longer schedule / larger encoder.** Lift the budget-imposed 500-epoch cap toward the 1000-epoch default and evaluate the ResEnc-XL planner, which the target 96 GB GPU can host.
4. **Learned post-processing thresholds.** Cross-validate the per-class component-size thresholds (Table 3) rather than fixing them, and consider a small classifier to gate rare-class predictions per case.
5. **Uncertainty and outlier flagging.** Use TTA/fold variance or an explicit uncertainty head to surface the long-tail outlier cases for human review — directly addressing the dominant source of rare-class error.

13 Conclusion (Summary)

In summary, this work delivers a complete, reproducible BraTS-PED 2026 Task-2 system: a skull-stripping front-end, a five-fold residual-encoder nnU-Net ensemble, and lesion-wise post-processing, packaged as a portable container and validated against ground truth at two evaluation scales. It reaches lesion-wise Dice of 0.797 (TC) and 0.834 (WT) in cross-validation and voxel-wise Dice of 0.876/0.914 (TC/WT) on a held-out internal test set, with a clearly characterised path to closing the gap to the leading 2025 systems. All artifacts are released to support reproducibility.

A Command Reference

The complete pipeline is driven by a handful of commands (see the repository `README.md` for full detail).

- **Environment:** `bash code/setup_env.sh` — creates `venv/`, installs the CUDA-12.8 PyTorch wheel and the `brats_ped` package and all dependencies.
- **Skull-strip:** `python code/skull_strip.py -split both -method auto -device gpu`.
- **Prepare + plan:** `python code/prepare_dataset.py; python code/train.py -step preprocess`.
- **Train (SLURM):** `bash code/slurm/submit_train.sh 0 1 2 3 4` — resource-aware submission, chained `prep→train`, resumable.
- **Infer:** `python code/inference.py -input data_ss/validation -output predictions/raw -raw-input -folds 0 1 2 3 4 -tta`.
- **Post-process:** `python code/postprocess.py -input predictions/raw -output predictions/postprocessed`.
- **Evaluate:** `python code/evaluate_lesionwise.py -pred predictions/postprocessed -gt <labelsTr> -output results/lesionwise.csv`.
- **Container:** `bash docker/build.sh` (stage model + build); `bash docker/run_sample.sh` (run + score on held-out cases).

B Output Format

The container reads `/input/<SubjectID>/` containing the four modalities (`<SubjectID>--{t1n,t1c,t2w,t2f}.n` and writes a single integer-valued label map `/output/<SubjectID>.nii.gz` with voxel values $\{0, 1, 2, 3, 4\} = \{\text{background, ET, NET, CC, ED}\}$, in the input geometry. This matches the BraTS-Lighthouse algorithm contract, so the same artifact serves the held-out evaluation and the official leaderboard submission.

Reproducibility

Code, training/inference scripts, SLURM job files, the container definition, the result CSVs and this report are available at https://github.com/ujjwalbaid0408/BraTS2026_PED. Install with `bash code/setup_env.sh`; train with `code/slurm/submit_train.sh 0 1 2 3 4`; run inference with `code/inference.py`; build the container with `docker/build.sh`; and score with `code/evaluate_lesionwise.py`. The held-out split is deterministic (seed 1337).

Data and Ethics Statement

This work uses the official MICCAI BraTS-PED 2026 dataset, which comprises de-identified, retrospectively collected multi-parametric MRI with expert reference annotations, distributed under the challenge’s data-use agreement via the Synapse platform. No new data were acquired and no identifiable patient information is used or released. The challenge data are *not* redistributed in our repository; only source code, derived metric tables and aggregate figures are released. Automated segmentation outputs from this system are intended as a research and expert-in-the-loop decision-support tool and are not a substitute for clinical judgement or a cleared medical device.

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